

Experiment # 12

Understanding of Residual Block

Objective/s:

- What is Residual block
- Implementation of Residual block
- Lab Tasks

Theory:

A residual block is a fundamental component of Deep Residual Networks (**ResNets**), which are designed to ease the training of very deep neural networks. The core idea behind a residual block is to allow the model to learn residual functions with reference to the layer inputs, instead of learning unreferenced functions. This helps in addressing the **vanishing gradient** problem and allows the training of deeper networks.

Structure of a Residual Block

A standard residual block consists of the following components:

1. **Input:** The input to the residual block.
2. **Convolutional Layers:** Two or more convolutional layers that learn to transform the input.
3. **Batch Normalization:** Applied after each convolution to normalize the activations.
4. **Activation Function:** Usually ReLU (Rectified Linear Unit) applied after each batch normalization.
5. **Shortcut Connection:** The original input is added to the output of the convolutional layers. This addition is what creates the "residual" connection.

FCN (fully conv.)

```
a = Conv2D(...) (input)
b = Conv2D(...) (a)
c = Conv2D(...) (b)
d = Conv2D(...) (c)
e = Conv2D(...) (d)
...
```

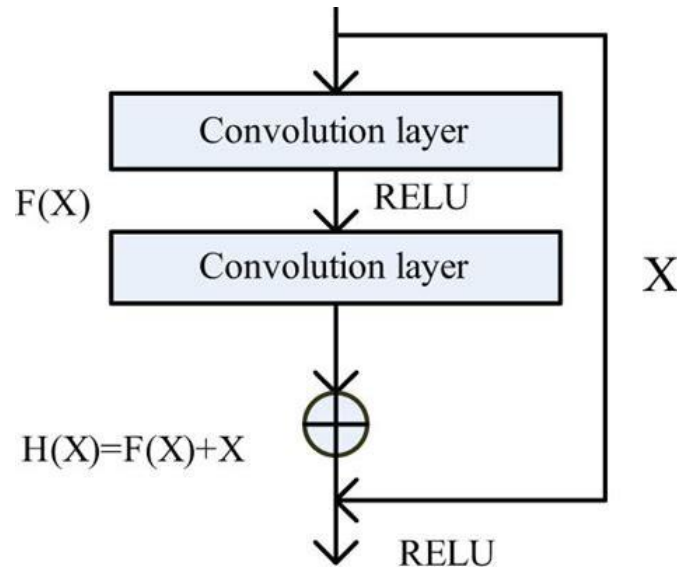
Residual

```
a = Conv2D(...) (input)
b = add(a, input)
c = Conv2D(...) (b)
d = add(c, b)
e = Conv2D(...) (d)
f = add(e, d)
...
```

Mathematical Representation

Let x be the input to the residual block, and $F(x)$ be the transformation applied by the convolutional layers within the block. The output y of the residual block is given by:

$$y = F(x) + x$$



Why Residual Blocks

▪ Learning the Difference

- Instead of learning the full function $H(x)$, it learns just the difference: $F(x) = H(x) - x$
- Then reconstructs as: $H(x) = F(x) + x$

▪ Gradient Flow

- The shortcut path helps gradients flow backward directly
- Avoids vanishing gradients in very deep networks

Lab Task

You must make residual block that has 70 layers and parameter <5M.

